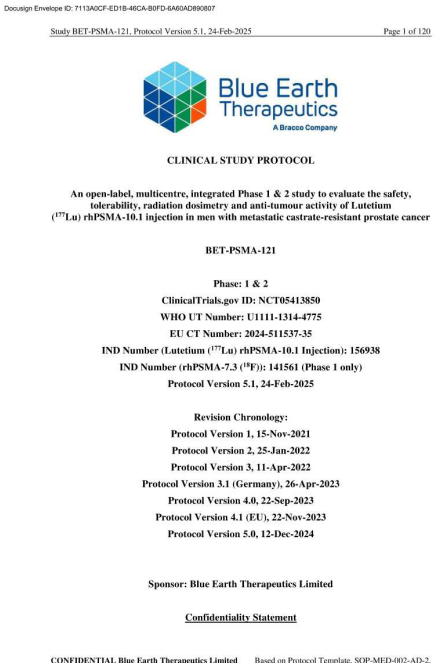
BET-121

^{177}Lu-rhPSMA-10.1 · integrated Phase 1 & 2 · NCT05413850

Blue Earth Therapeutics · P01 in Board Studio · 120 pages



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Blue Earth Therapeutics
A Bracco Company

CLINICAL STUDY PROTOCOL

An open-label, multicentre, integrated Phase 1 & 2 study to evaluate the safety, tolerability, radiation dosimetry and anti-tumour activity of Lutetium (^{177}Lu) rhPSMA-10.1 injection in men with metastatic castrate-resistant prostate cancer

BET-PSMA-121

Phase: 1 & 2
ClinicalTrials.gov ID: NCT05413850
WHO UT Number: U1111-1314-4775
EU CT Number: 2024-511537-35
IND Number (Lutetium (^{177}Lu) rhPSMA-10.1 Injection): 156938
IND Number (rhPSMA-7.3 (^{19}F)): 141561 (Phase 1 only)
Protocol Version 5.1, 24-Feb-2025

Revision Chronology:
Protocol Version 1, 15-Nov-2021
Protocol Version 2, 25-Jan-2022
Protocol Version 3, 11-Apr-2022
Protocol Version 3.1 (Germany), 26-Apr-2023
Protocol Version 4.0, 22-Sep-2023
Protocol Version 4.1 (EU), 22-Nov-2023
Protocol Version 5.0, 12-Dec-2024

Sponsor: Blue Earth Therapeutics Limited

Confidentiality Statement

CONFIDENTIAL Blue Earth Theranetics Limited Based on Protocol Template: SOP-MED-I002-AD-2.

What this study is, in one breath

Give lutetium-177 radiohybrid PSMA therapy to PSMA-positive mCRPC, watch PSA, and stop if kidney or marrow dose caps are hit on SPECT/CT.

This is radioligand therapy, not a diagnostic PET trial.

If the protocol feels like a wall of words

A protocol is a recipe: who can join, what is injected, how much, when you scan or treat, and how you decide if a picture or outcome was right. This sheet is the same idea in normal English. It does not replace the signed site copy.

1. The drug

BET-121 is ^{177}Lu-rhPSMA-10.1 (Blue Earth Therapeutics, NCT05413850): a radiohybrid small-molecule PSMA ligand for radioligand therapy in PSMA-positive mCRPC. Same targeting idea as Pluvicto, different ligand pharmacokinetics (faster kidney washout in mice than 7.3C or PSMA-I&T).

- Lu-177: T½ 6.7 days, β− therapy. Activity tolerance ±10% of prescribed.
- Extravasation = SAE. Large-bore IV, saline flush.
- Kidney is the dose-limiting organ. SPECT/CT uses the therapeutic emission — no extra tracer.

2. Design

Part	Who	Dose idea
Phase 1 (done)	Post-taxane mCRPC · n ≈ 12 · BOIN	5.55 then 7.40 GBq q6w, max 3 cycles
Phase 2A	Chemo-naïve for mCRPC · n ≈ 30	2 × 10.00 GBq then ≤5 × 7.40 GBq, all q6w
Phase 2B	Same pool · n ≈ 30	≤8 × 7.40 GBq; first 3 at q3w then q6w
Phase 2C optional	TSC may open · n ≈ 10	2 × 14.80 GBq then ≤4 × 7.40 q6w

Phase 2 is open-label, ~70 people, ~25 centres. Primary: PSA50 (≥50% PSA drop from baseline to end of treatment). Futility if estimated PSA50 ≤ 20% in a cohort.

3. Organ limits and SPECT clock

- Kidney 23 Gy (Phase 2 absolute 35 Gy, re-consent if exceeding 23). Marrow 2 Gy (absolute 2.5). Salivary 24 Gy tracked. eGFR ≥ 60 (CKD-EPI 2009).
- Phase 1 SPECT every cycle at 3, 24, 48, 168 h + PK blood.
- Phase 2: Cycle 1 at 3/24/48/168 h; later cycles a single 48 h scan (2B first two keep 168 h). Results back before the next cycle so caps can stop dosing.

Do not mix this with diagnostic PSMA PET

POSLUMA (^{18}F-rhPSMA) 296 MBq is mentioned for Phase 1 imaging only. The therapy molecule is ^{177}Lu-rhPSMA-10.1.

What to know cold

Q. Ligand?
A. rhPSMA-10.1. Same PSMA idea as Pluvicto, different PK (faster kidney washout).

Q. Nuclide?
A. Lu-177, T½ 6.7 d, β−. ±10% of prescribed. Extravasation = SAE.

Q. Phase 2 primary?
A. PSA50 from baseline to end of treatment.

Q. Kidney cap?
A. 23 Gy (Phase 2 absolute 35 Gy). Marrow 2 Gy (absolute 2.5).

Q. SPECT?
A. Uses the therapeutic emission. Results back before the next cycle.

How to use this with Board Studio

Open the matching protocol module and press play. The left pane is the signed page. Use this PDF when legalese stalls you. If anything conflicts, the signed protocol wins.

Disclaimer. Educational summary for board-style study. Not medical advice, not a recruitment brochure, not a substitute for the signed protocol, and not affiliated with the sponsor, ABR, ABNM, ACGME, SNMMI, NRC, or FDA. Confirm numbers against the current site copy before any clinical use.